REAL-WORLD SOFTWARE CASE STUDY

GYM MANAGEMENT SYSTEM

Software Engineering Case Study

Prepared by: K Dheeran
Student ID: 2420090149
B.Tech CSE / IT
KL University

1. Introduction

The Gym Management System is a web-based application designed to manage the daily activities of a gym in one place. It can be used by the administrator, trainers and members. In a traditional gym, member records, payments, attendance and workout details may be maintained using registers or separate files. This can make searching and updating information difficult.

The proposed system stores these details digitally and provides different functions according to the user's role. An administrator can manage members, trainers, membership plans and payments. Trainers can manage workout plans, while members can view their membership, attendance and workout information. The main objective is to reduce manual work and make gym operations easier to manage.

2. User Stories

User	User Story	Purpose
Admin	As an admin, I want to register members so that their details are stored digitally.	Member management
Admin	As an admin, I want to manage membership plans so that available plans stay updated.	Plan management
Admin	As an admin, I want to record payments so that transactions can be tracked.	Payment tracking
Admin	As an admin, I want to generate reports so that I can monitor gym operations.	Reporting
Trainer	As a trainer, I want to create workout plans so that members receive suitable exercises.	Workout management
Trainer	As a trainer, I want to view member details so that I can guide them properly.	Member support
Member	As a member, I want to view my	Membership tracking

	membership details so that I know its validity.	
Member	As a member, I want to view my attendance so that I can track my visits.	Attendance tracking
Member	As a member, I want to view my workout plan so that I can follow my exercises.	Workout access

3. Software Development Process Model

The Agile software development model is selected for this application. The system can be developed in small iterations, and feedback can be collected after each iteration. This is useful because gym requirements may change after the administrator and trainers start using the system.

- Iteration 1: Login, user roles and member registration.
- Iteration 2: Membership plans and payment management.
- Iteration 3: Trainer management and attendance.
- Iteration 4: Workout plans and member dashboard.
- Iteration 5: Testing, bug fixing and deployment.

4. Programming Language and Technologies

Technology	Choice	Purpose
Frontend	HTML, CSS, JavaScript / React	Build the user interface
Backend	Java with Spring Boot	Implement business logic and REST APIs
Database	MySQL	Store member, payment and attendance data
API	REST API	Connect frontend and backend
Testing	JUnit and Postman	Test application functions and APIs
Version Control	Git and GitHub	Track source code changes
Deployment	AWS / cloud hosting	Make the application available online

5. Implementation Strategy

5.1 Requirement Analysis

Identify the requirements of the admin, trainer and member. Define the main features and access permissions.

5.2 Database Design

Create tables for members, trainers, memberships, payments, attendance and workout plans. Define suitable relationships between them.

5.3 Backend Development

Develop REST APIs using Spring Boot. Add validation, authentication and role-based access.

5.4 Frontend Development

Create login, dashboard, member management, payment, attendance and workout-plan screens.

5.5 Integration

Connect the frontend with backend APIs and connect the backend with the MySQL database.

5.6 Deployment

Test the complete application and deploy it on a cloud server. Regular backups can be maintained for important data.

6. Evaluation Strategy

Evaluation Area	Approach
Functional Testing	Check whether registration, login, membership, payments, attendance and workout functions work correctly.
Performance Testing	Measure response time and check the application with multiple users performing common operations.
Security Testing	Verify authentication, authorization, password protection and prevention of unauthorized access.
Usability Testing	Ask a small group of users to perform common tasks and collect feedback about ease of use.
Acceptance Testing	Compare the completed system with the original requirements and verify that the major user stories are satisfied.

7. UML Diagrams

7.1 Use Case Diagram

The use case diagram shows the main interactions between the three users of the system and the functions they can access.

7.2 Class Diagram

The class diagram represents the main data objects of the system and their relationships.

8. Conclusion

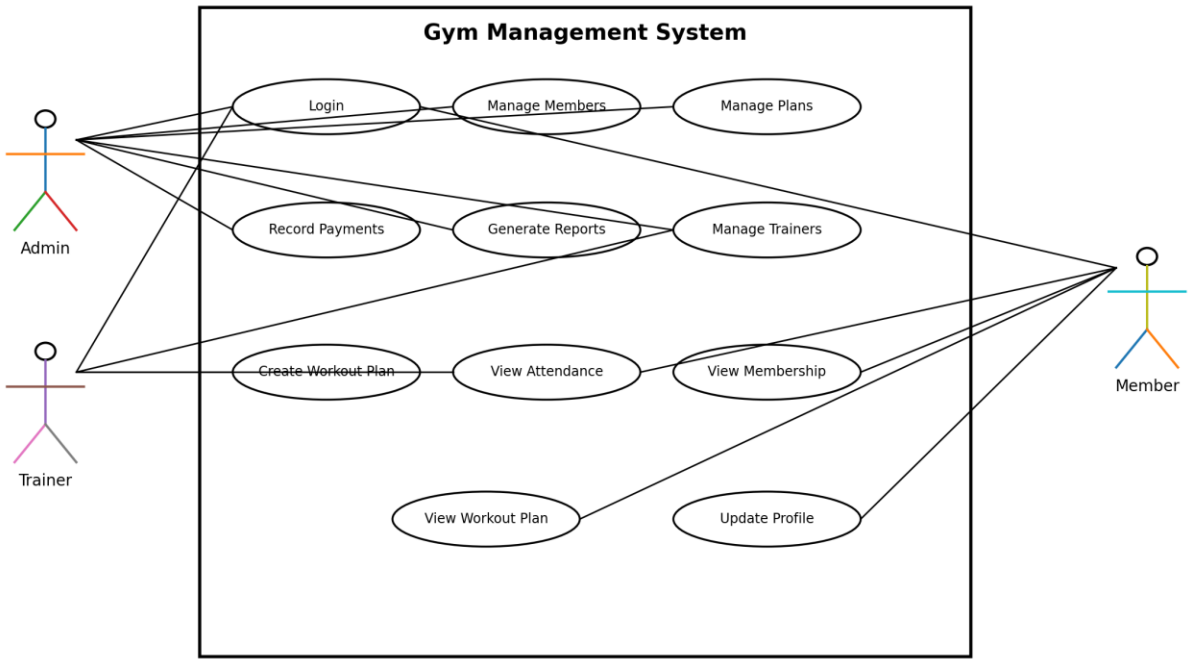
The Gym Management System provides a practical solution for managing common gym activities digitally. It brings member records, memberships, payments, attendance and workout plans into one application. Using Agile development makes it possible to build the system in smaller iterations and improve it using feedback. The proposed technologies are suitable for developing a maintainable web application, and the system can later be extended with features such as QR-based attendance, payment notifications, mobile support and online membership renewal.

End of Case Study

Corrected UML Diagrams

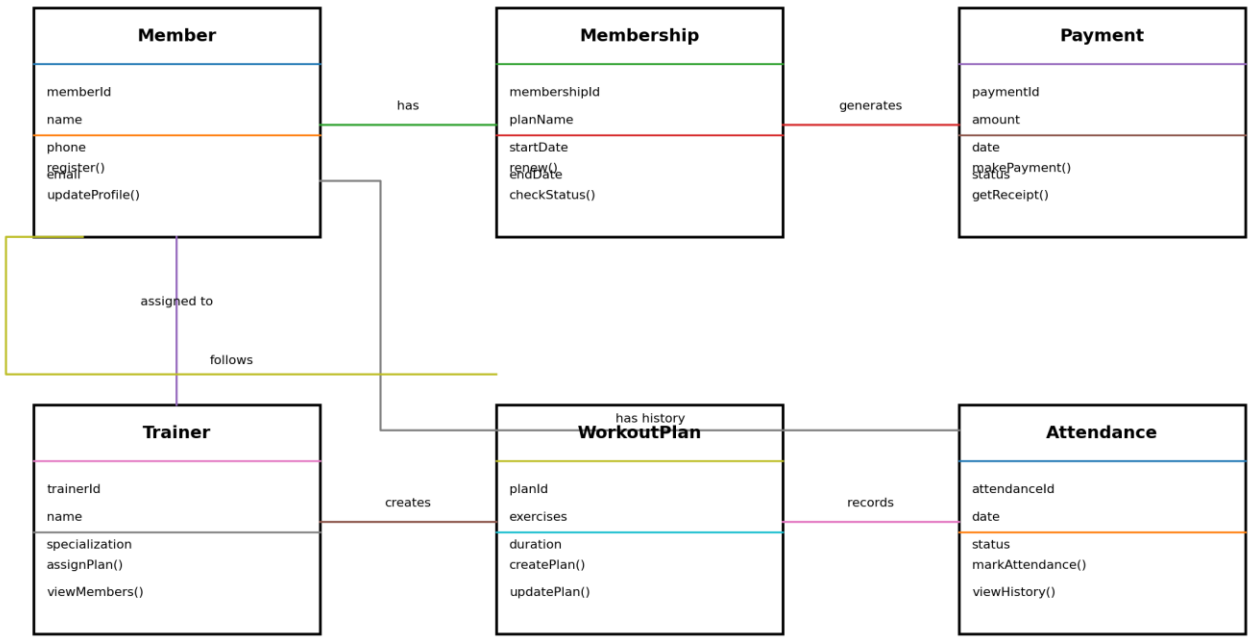
The following diagrams use direct, clearly routed associations so each relationship is visually connected to the intended actor or class.

Use Case Diagram



Use Case Diagram

Class Diagram



Class Diagram